Wireless Standards and Configuration

Setup, Channels, Security, MAC Filtering and ACLs

1. Theory Summary

- SSID: the network name that the router broadcasts.
- Frequency bands: 2.4 GHz (longer range, more interference, only channels 1, 6 and 11 do not overlap) and 5 GHz (faster, shorter range, many non-overlapping channels).
- Channel selection: neighbouring networks on the same or overlapping channels cause interference. Pick the channel with the fewest networks and least overlap.
- Security: WPA3 is the strongest, WPA2 with AES (CCMP) is the minimum recommended. Avoid WEP, WPA and TKIP, and turn off WPS.
- MAC filtering: allows only listed device hardware addresses. It is a basic access control, but MAC addresses can be spoofed, so it must not replace strong encryption.
- ACL: a list of permit or deny rules that filters traffic by IP address, protocol or port. Domain names are not matched directly by classic ACLs.

2. Task 1: Basic Wireless Setup

Steps:

- Power on the router and open its admin page in a browser (commonly 192.168.0.1 or 192.168.1.1; check the label under the router).
- Log in and open the Wireless settings. Note the SSID and channel.
- Connect your laptop or phone to the SSID and confirm internet access.

Item	Your observation
Router model	
SSID	
Band (2.4 GHz or 5 GHz)	
Default channel / mode	(for example Auto, channel 6)
Default security	(for example WPA2-PSK)
Connected device	(laptop or phone)

3. Task 2: Select a Manual Channel and Check Interference

Steps:

- Install a Wi-Fi analyzer app (for example WiFiman, NetSpot or Wifi Analyzer on Android).

- Open the channel graph on the 2.4 GHz band and note how many networks sit on channels 1, 6 and 11 and how strongly they overlap.
- In the router's wireless settings, change Channel from Auto to the least crowded of 1, 6 or 11, and save.
- Reconnect, rescan with the app, and compare the graph before and after.

Channel	Networks seen (before)	Networks seen (after)
1		
6		
11		

Tip: channels 1, 6 and 11 are 25 MHz apart, so they do not overlap each other. A network on channel 3, for example, overlaps both 1 and 6. If the 5 GHz band is available and your devices support it, it usually has far less interference.

4. Task 3: Enable WPA2 or WPA3 Security

Steps:

- Open Wireless Security settings and select WPA3 (or WPA2-PSK with AES if WPA3 is not available, or WPA2/WPA3 mixed mode for older devices).
- Set a strong passphrase: at least 12 characters, mixing upper and lower case, numbers and symbols, and not related to personal details.
- Also change the router admin password and disable WPS.
- Forget the old network on your device and reconnect using the new password. Try a wrong password once to confirm that it is rejected.

Check	Result
Security mode set	
Connected with the new password?	
Wrong password rejected?	

5. Task 4: MAC Filtering

Finding the MAC address:

- Android: Settings > About phone > Status > Wi-Fi MAC address.
- iPhone: Settings > Wi-Fi > the (i) next to your network > Private Wi-Fi Address.
- Windows: open Command Prompt and run ipconfig /all, then look for Physical Address of the Wi-Fi adapter.

Note: modern phones use a random (private) MAC address per network. Either turn off the private or random MAC for this network, or add the address shown for this network to the list.

Steps:

- In the router, open Wireless > MAC Filtering (or Access Control).
- Set the mode to Allow only the listed devices and add your device's MAC address.
- Connect with the listed device (should succeed), then try a second device that is not on the list (should fail).

Device	MAC address	Connection result
Allowed device		
Other device		

Observation: the unlisted device cannot connect even with the correct Wi-Fi password. Limitation: an attacker can copy a listed MAC address, so MAC filtering is not strong security by itself.

6. Task 5: ACL to Block a Website

Option A: home router. Most home routers do not let you write an ACL directly. Use Access Restrictions, Parental Controls or Website Blocking, add www.youtube.com (and youtube.com) as a blocked domain, and apply it to all devices.

Option B: Cisco router (lab or Packet Tracer). A standard IOS ACL matches IP addresses, not names, so first find the website's address:

```
C:\> nslookup www.youtube.com
```

Then block it on the router interface facing the Wi-Fi/LAN network (replace 203.0.113.10 with the address returned, and repeat for each address):

```
Router(config)# ip access-list extended BLOCK-YOUTUBE
Router(config-ext-nacl)# deny ip 192.168.1.0 0.0.0.255 host 203.0.113.10
Router(config-ext-nacl)# permit ip any any
Router(config-ext-nacl)# exit
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip access-group BLOCK-YOUTUBE in
Router(config-if)# end
Router# show access-lists
```

Notes:

- The final permit ip any any is needed, because every ACL ends with an implicit deny all.
- The in direction on the LAN-facing interface filters traffic as it enters from the Wi-Fi network.
- Large sites use many changing IP addresses and also QUIC (UDP 443). Domain-based website blocking, or a DNS filter, is more reliable for real use.

Testing:

- From a connected device, open www.youtube.com in a browser. It should not load (timeout or blocked page).

- Open another site such as www.google.com. It should still load, which shows the ACL is blocking only the target.
- If the site still opens, clear the browser cache and run ipconfig /flushdns, then check show access-lists for hit counts.

Test	Result
www.youtube.com from a Wi-Fi device	
Another website (for example google.com)	

7. Troubleshooting

Problem	Check
Cannot connect after changing the channel	Rescan Wi-Fi; some old devices do not support channels 12 to 13
Cannot connect after changing security	Forget the network on the device and reconnect
Allowed device is blocked by MAC filtering	Use the correct (not randomized) MAC address
Website still opens after the block	Flush DNS, clear cache, check all resolved IPs and the ACL direction

8. Conclusion

A basic wireless network was set up, the channel was changed manually to reduce interference, WPA2 or WPA3 was enabled with a strong password, MAC filtering restricted access to a chosen device, and an ACL or website block was used to stop access to a specific website. Layering these controls gives better protection than any single one.

9. Submission Checklist

- Screenshot of router wireless settings (SSID, default channel)
- Analyzer screenshots before and after the channel change
- Screenshot of the security mode (WPA2 or WPA3)
- MAC filter list and the result for the unlisted device
- ACL or website block settings and the blocked-site test result